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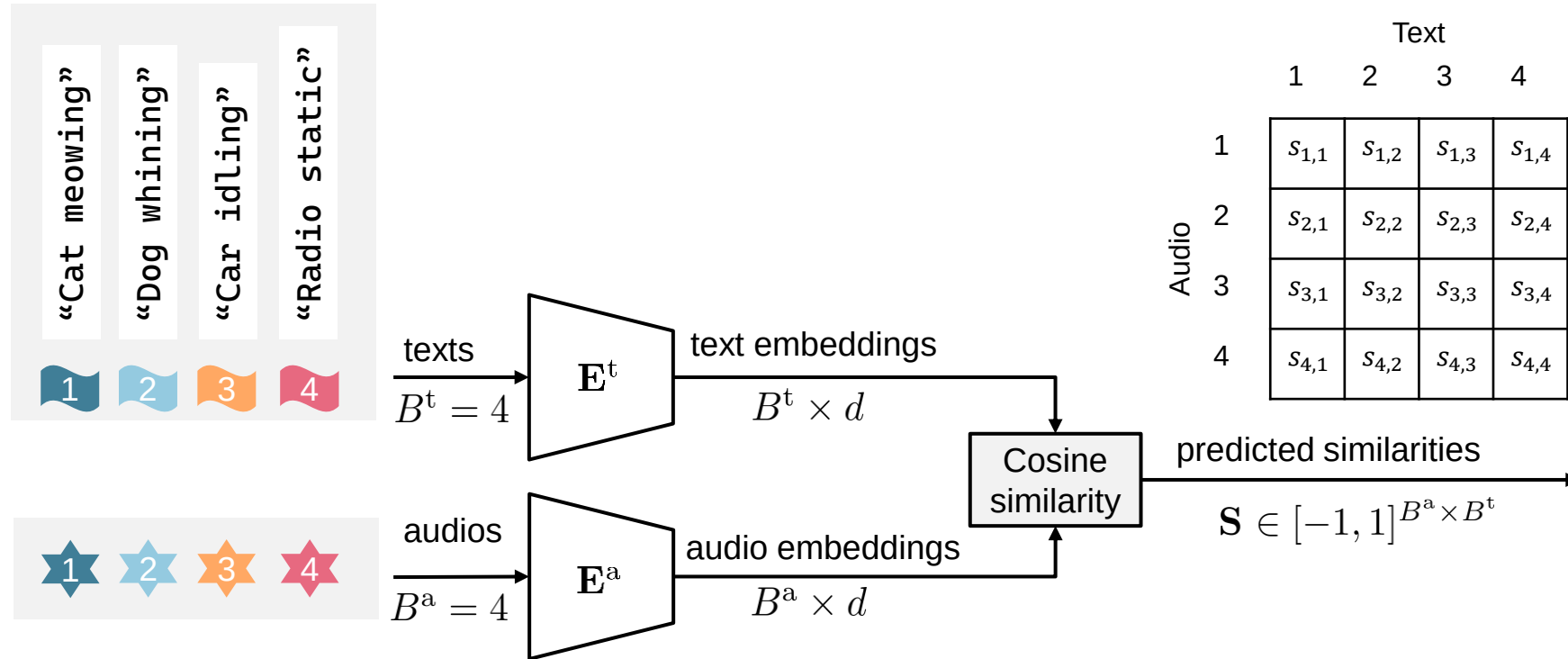
Leveraging Mutual Intra-Modal Similarity Supervision for Text and Audio

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CLAP – Contrastive Language-Audio Pre-Training

Inference

Audio-language models can compute similarity scores between texts and audios.



Elizalde et al., "CLAP: Learning Audio Concepts From Natural Language Supervision," 2023.

CLAP – Contrastive Language-Audio Pre-Training

Applications

The predicted similarities can easily be used to solve three different tasks.

		Text			
		1	2	3	4
Audio	1	$s_{1,1}$	$s_{1,2}$	$s_{1,3}$	$s_{1,4}$
	2	$s_{2,1}$	$s_{2,2}$	$s_{2,3}$	$s_{2,4}$
	3	$s_{3,1}$	$s_{3,2}$	$s_{3,3}$	$s_{3,4}$
	4	$s_{4,1}$	$s_{4,2}$	$s_{4,3}$	$s_{4,4}$

predicted similarities

$$\mathbf{S} \in [-1, 1]^{B^a \times B^t}$$

top-1 per row

Zero-Shot Classification (ZSC)

For each audio, which text
“label” does it belong to?

Metric: Accuracy

top-K per row

Audio-queried Text Retrieval (ATR)

For each audio, which texts are the K most
relevant ones in a bigger text corpus?

Metrics: mAP@K, Recall@K

top-K per column

Text-queried Audio Retrieval (TAR)

For each text, which audios are the K most
relevant ones in a bigger audio corpus?

Metrics: mAP@K, Recall@K

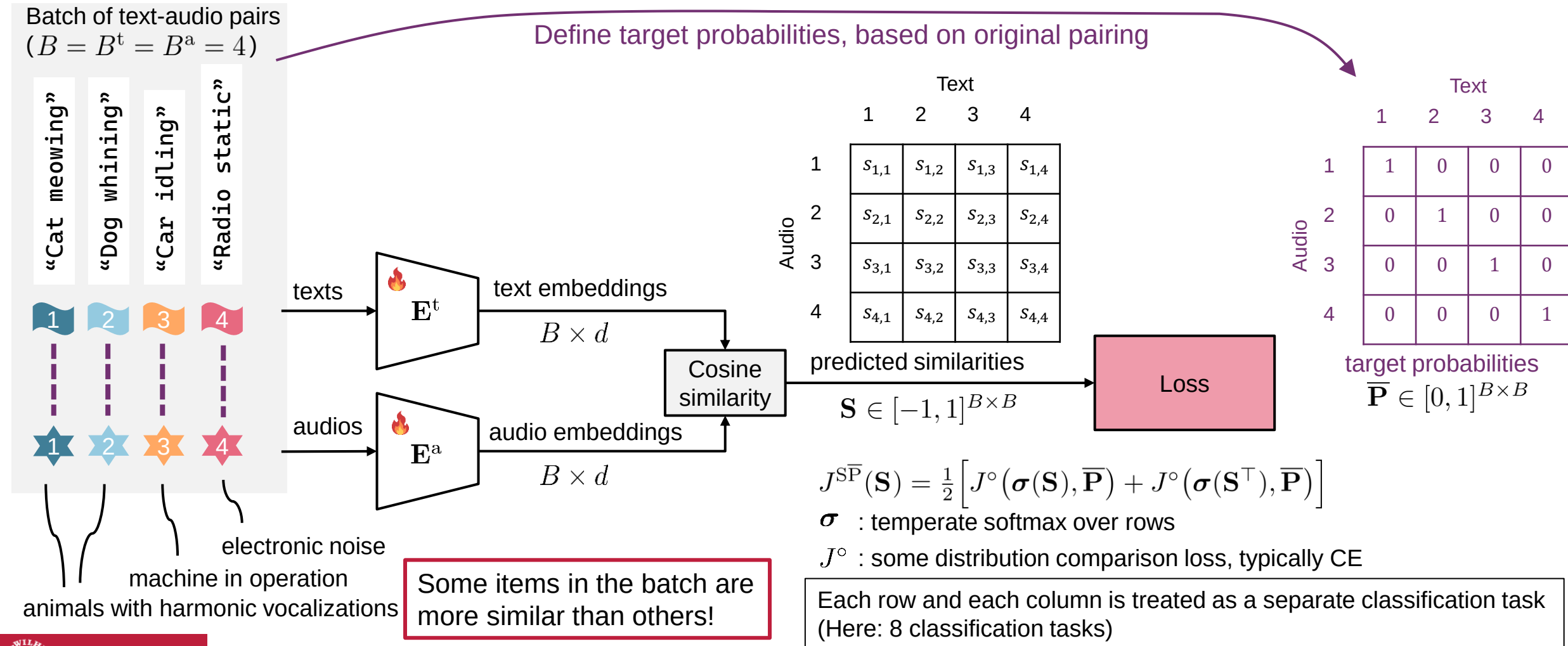
Elizalde et al., “CLAP: Learning Audio Concepts From Natural Language Supervision,” 2023.

CLAP – Contrastive Language-Audio Pre-Training

Training

The CLAP-style training setup is used in most text-audio methods.

Elizalde et al., “CLAP: Learning Audio Concepts From Natural Language Supervision,” 2023.



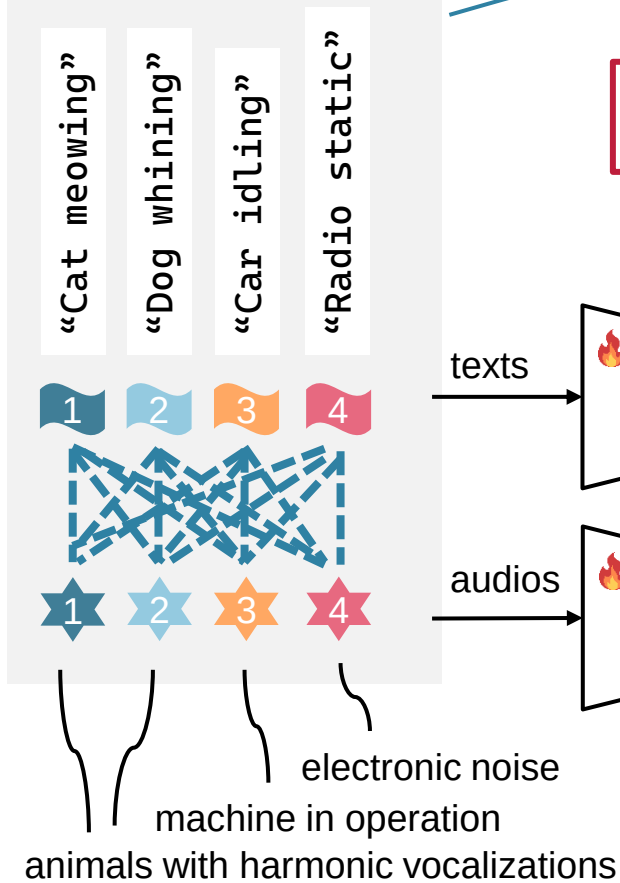
CLAP – Contrastive Language-Audio Pre-Training

Towards Training with Soft Targets

The CLAP-style training setup is used in most text-audio methods.

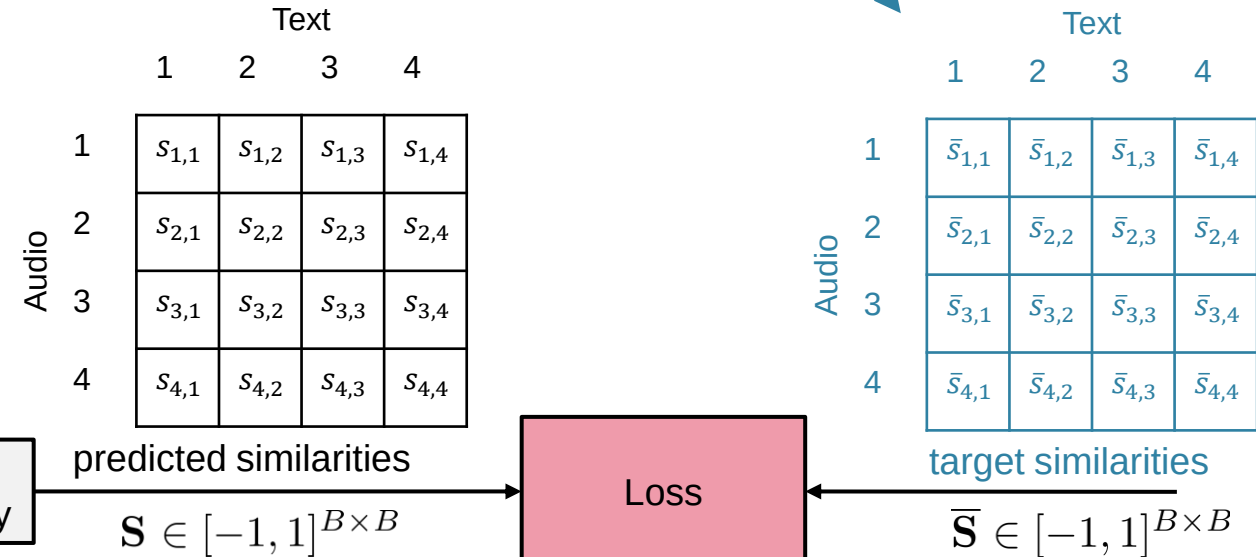
Sobal et al., “X-Sample Contrastive Loss: Improving Contrastive Learning with Sample Similarity Graphs,” 2025.

Batch of text-audio pairs
($B = B^t = B^a = 4$)



We mostly deal with the construction of training targets.

Define target similarities, considering all batch relations



$$J^{\bar{S}}(S, \bar{S}) = \frac{1}{2} \left[J^{\circ}(\sigma(S), \bar{\sigma}(\bar{S})) + J^{\circ}(\sigma(S^T), \bar{\sigma}(\bar{S})) \right]$$

$\sigma, \bar{\sigma}$: temperate softmax over rows, using different temperatures

J° : some loss that compares distributions, **CE** or **MSE**

Each row and each column is treated as a separate classification task
(Here: 8 classification tasks)

Our Proposals

Overview

- 1) Bring Sobal et al. [1] from the text-image into the text-audio field
- 2) Generalize Sobal et al. by making use of each branch's intra-modal information
- 3) Use a mixing process informed by embedding uncertainties
- 4) Test an optional modality classification branch to minimize the modality gap

We use a baseline by Primus et al. [2] with publicly available data and pre-trained encoders.

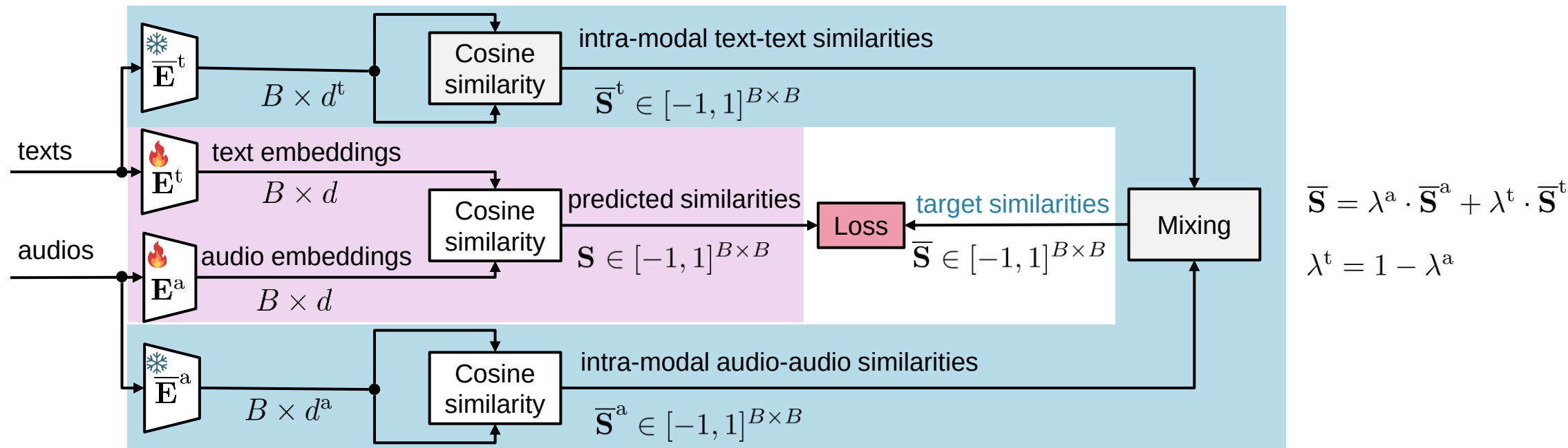
[1] Sobal et al., "X-Sample Contrastive Loss: Improving Contrastive Learning with Sample Similarity Graphs," 2025.

[2] Primus et al., "Estimated Audio-Caption Correspondences Improve Language-Based Audio Retrieval," 2024.

Our Proposals

Intra-Modal Similarities with Simple Mixing (+T/+A/+T+A)

Use frozen encoders to predict intra-modal similarities on the text and the audio side (generalization of [1]).
Final target similarities are a weighted average of the intra-modal similarities.



$\lambda^t = 1, \lambda^a = 0$ Only use text-text intra-modal similarities as targets. (+T)

$\lambda^t = 0, \lambda^a = 1$ Only use audio-audio intra-modal similarities as targets. (+A)

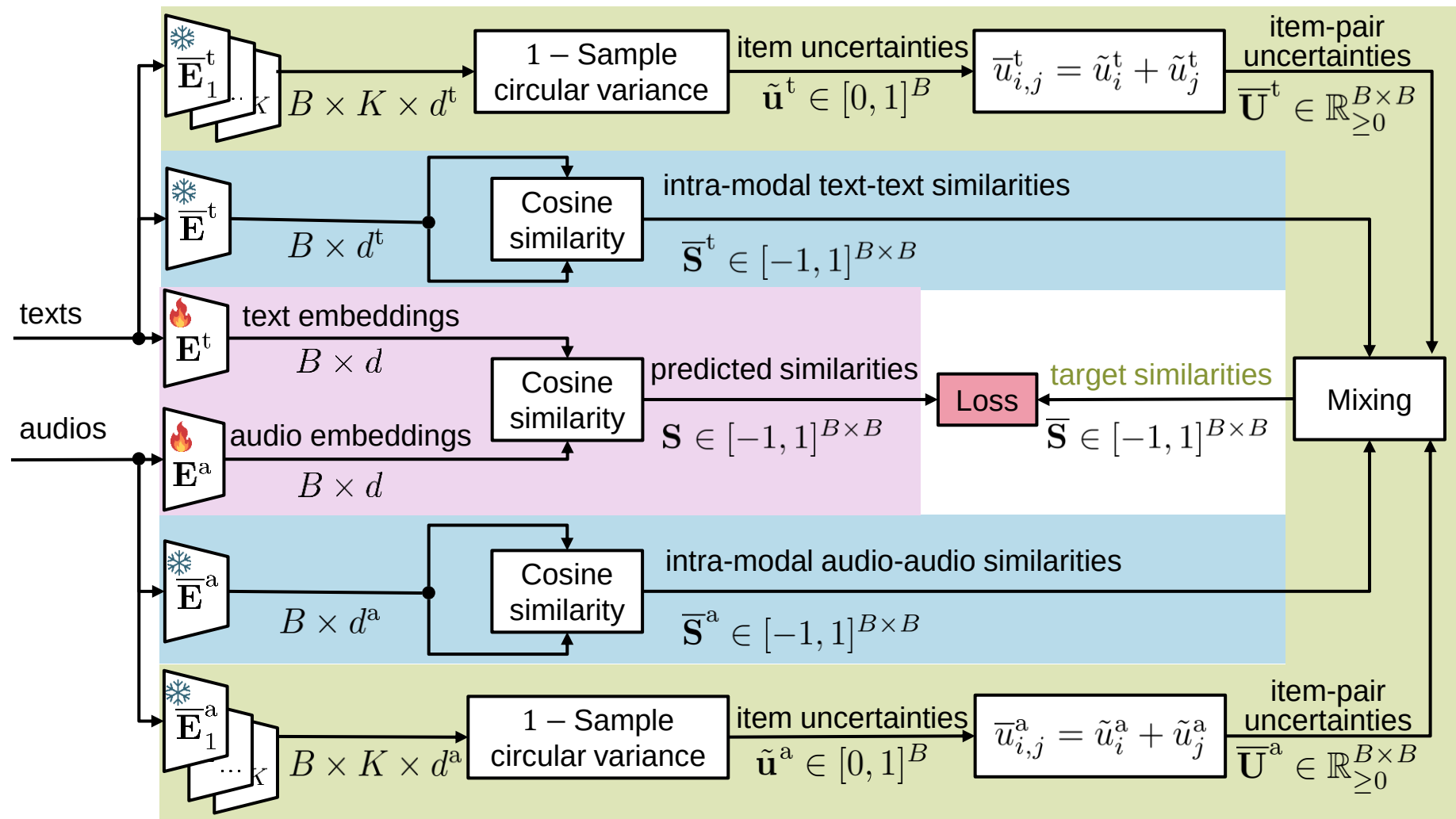
$\lambda^t = 0.5, \lambda^a = 0.5$ Use text-text and audio-audio intra-modal similarities equally for targets. (+T+A)

[1] Sobal et al., "X-Sample Contrastive Loss: Improving Contrastive Learning with Sample Similarity Graphs," 2025.

Our Proposals

Improved Mixing with Target Uncertainty Estimation (+T+A+U)

Inform the mixing process by how uncertain the embeddings of $\bar{\mathbf{E}}^t$ and $\bar{\mathbf{E}}^a$ are, using a Monte Carlo approach. We perturb the models $\bar{\mathbf{E}}^t$ and $\bar{\mathbf{E}}^a$ K times, using dropout layers.



Final target similarities are element-wise weighted averages.

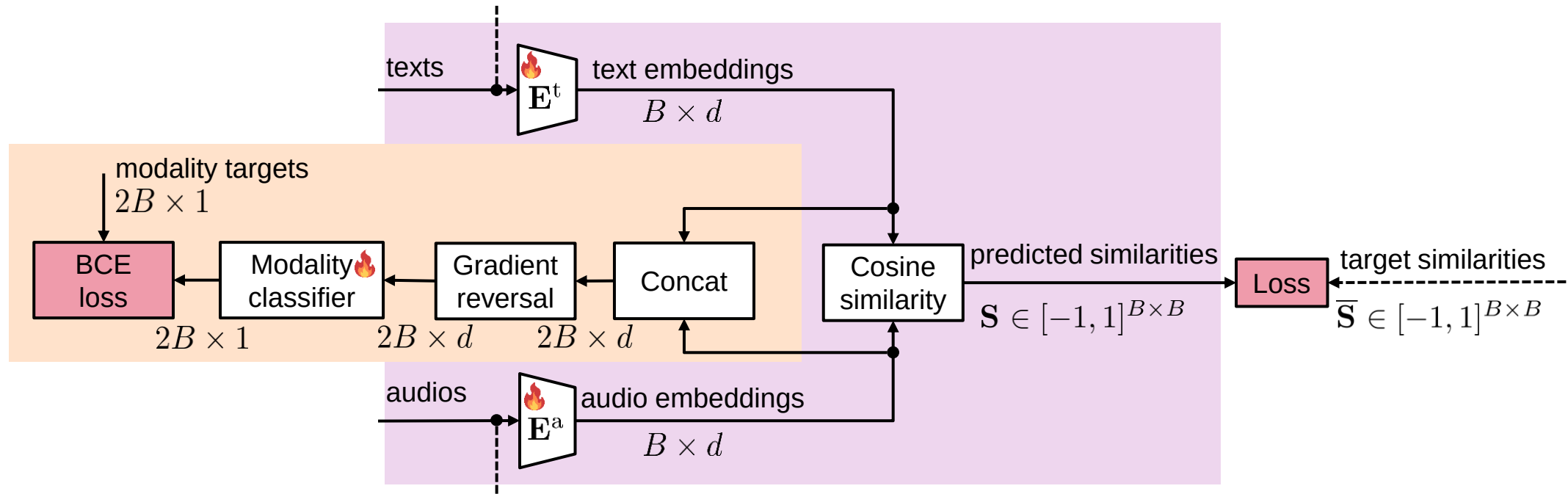
$$\bar{s}_{i,j} = w(\bar{u}_{i,j}^t, \bar{u}_{i,j}^a) \cdot \bar{s}_{i,j}^a + w(\bar{u}_{i,j}^a, \bar{u}_{i,j}^t) \cdot \bar{s}_{i,j}^t$$

$$w(a, b) = \begin{cases} \frac{a}{a+b}, & a + b > 0 \\ 0.5, & a + b = 0 \end{cases}$$

Our Proposals

Ablation: Modality Classification (+M)

We also test adding a modality classification branch with a gradient reversal layer. The goal is to make the modalities of embeddings indistinguishable.



Experiments

Results

- [1] Elizalde et al., “Natural Language Supervision for General-Purpose Audio Representations,” 2024.
 [2] Primus et al., “Estimated Audio-Caption Correspondences Improve Language-Based Audio Retrieval,” 2024.
 [3] Mei et al., “WavCaps: A ChatGPT-Assisted Weakly-Labelled Audio Captioning Dataset for Audio-Language Multimodal Research,” 2024.
 [4] Kim et al., “AISTAT Lab System for DCASE 2025 Task 6: Language-Based Audio Retrieval,” 2025.
 [5] Sofer et al., “Pull It Together: Reducing the Modality Gap in Contrastive Learning,” 2025.

Batch size		various	1, 536	64																
#training samples		various	4.6M	0.5M																
Task	Test Dataset	Metric (%)	SOTA*		CLAP23 [1]		Baseline [2]		(ours)										MSE+T+A+U+M	
									CE+T	CE+A	CE+T+A	CE+T+A+M	CE+T+A+U	CE+T+A+U+M	MSE+T	MSE+A	MSE+T+A	MSE+T+A+M		
ZSC	ESC50	ACC	94.8	[3]	94.1	95.9	95.1	95.2	95.3	95.9	96.1	95.3	96.5	94.1	95.2	95.8	95.8	96.2		
	TUT17	ACC	53.8	[1]	55.5	57.8	64.0	58.8	56.5	59.4	57.9	60.7	62.0	60.1	58.3	61.7	58.3	55.6		
	US8K	ACC	82.3	[1]	82.6	80.1	82.3	81.3	81.8	83.2	82.3	82.5	83.4	82.6	82.4	80.4	80.0	81.1		
	NSynth Instr.	ACC	47.9	[1]	44.7	33.1	34.7	34.0	32.8	33.0	34.7	34.5	32.4	35.4	34.5	33.2	35.7	36.3		
	Average rank ↓		—		6.5	8.5	5.0	7.3	8.8	5.5	5.8	5.0	4.0	5.3	6.5	6.5	7.0	6.3		
TAR	Clotho-eval	mAP@10	41.7	[4]	25.8	32.3	33.6	33.2	32.6	32.5	32.8	33.7	34.4	34.4	34.4	34.3	34.3	34.2		
	Clotho-eval	amAP@16	48.8	[4]	29.4	38.5	39.0	38.9	38.6	39.1	39.6	40.0	39.7	39.1	39.1	39.6	40.3	39.7		
	Clotho-eval	R@1	28.4	[4]	16.0	21.3	22.4	21.8	21.2	21.2	21.3	22.1	22.7	22.8	22.7	22.7	22.7	22.7		
	Average rank ↓		—		9.7	8.3	4.7	6.0	7.7	7.0	6.3	3.3	2.0	2.3	2.7	2.0	1.7	2.7		
ATR	Clotho-eval	mAP@10	15.5	[1]	15.6	19.8	19.2	19.2	18.5	19.1	19.4	18.3	20.7	20.1	20.8	20.7	19.5	20.0		
	Clotho-eval	R@1	26.6	[5]	22.9	29.0	26.5	27.4	24.7	24.9	26.7	25.7	29.1	28.3	30.7	28.4	27.3	27.6		
	Average rank ↓		—		13.0	4.0	9.0	7.5	11.5	10.5	8.0	11.0	2.0	4.0	1.0	3.0	7.0	5.0		

Less training complexity than CLAP23 [1].

No method excels at all tasks, but our variants dominate each task.

Intra-modal audio-audio targets perform quite well, overall.

Three of our variants consistently match or excel the baseline and CLAP23 for every task.

CE vs. MSE: MSE generally excels.

Conclusions

For a CLAP-style training, creating **good targets** is essential to performance on downstream tasks.

- Simple **intra-modal similarities**, produced by an expert model, lead to **consistent performance improvements**.
- Intra-modal **audio-audio targets are valuable** on their own, contrary to intuition.
- By exploiting **uncertainty estimations**, we further **improve on the text-queried audio retrieval (TAR) task**.
- Computationally demanding parts of our target estimation **can be computed offline** before training.
- These methods **can be used for any multi-modality contrastive training**.

Surprisingly, we show that **MSE quite often outperforms CE** as base loss.

- We hypothesize that MSE stabilizes training at smaller batch sizes better.

The **modality classification branch** does not necessarily lead to improved performance.

Using our method, **we excel CLAP23 [1] on every task, while using a fraction of data and batch size during training.**

[1] Elizalde et al., “Natural Language Supervision for General-Purpose Audio Representations,” 2024.

Thank you for your attention.

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